

1. Administrative & Study Identification

Full Study Title: Study Assessing the Efficacy and Safety of Alpelisib + Nab-paclitaxel in Subjects With Advanced TNBC Who Carry Either a PIK3CA Mutation or Have PTEN Loss **Short Title/ Acronym:** EPIK-B3 **Protocol Number:** CBYL719H12301 **NCT Number:** NCT04251533 **Posting ID:** 996660315249619740 **Sponsor/Company Name:** Novartis Pharmaceuticals **Investigational Product:** Alpelisib (BYL719) / Placebo + Nab-paclitaxel **Phase of Development:** Phase III **Trial Status:** Active, not recruiting **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine whether treatment with alpelisib in combination with nab-paclitaxel is safe and effective in subjects with advanced triple-negative breast cancer (aTNBC). Specifically, to evaluate Progression-Free Survival (PFS) in subjects who carry a PIK3CA mutation (Study Part A) and to evaluate confirmed Overall Response Rate (ORR) in subjects with PTEN loss (Study Part B1). **Secondary Objectives:** To evaluate Overall Survival (OS) in Part A, Objective Response Rate (ORR) with confirmed response in Part A, Clinical Benefit Rate (CBR) with confirmed response in Part A, Duration of Response (DOR), Time to Response (TTR), and to assess safety, tolerability, and pharmacokinetics.

2.2 Background & Rationale Triple-negative breast cancer (TNBC) is an aggressive subtype of breast cancer. Mutations and deregulations in the phosphatidylinositol-3-kinase (PI3K) pathway are common in breast cancer and promote tumor growth, progression, and resistance to standard chemotherapies like taxanes. Activating mutations in PIK3CA are noted in approximately 8-10% of TNBC patients. This study evaluates whether targeting the PI3K-alpha isoform with alpelisib, alongside standard chemotherapy (nab-paclitaxel), can overcome resistance and extend survival in this genetically defined patient population across multiple study parts (Part A, Part B1, and Part B2).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (in Part A) **Blinding:** Double-blind (in Part A) **Randomization:** Randomized (1:1 in Part A)

3.2 Planned Enrollment & Duration Target **\$N\$: 137 Number of Sites:** Multicenter (Global) **Status Details:** The trial status is Active, not recruiting. The recruitment of Part A was halted on 11-Nov-2022 due to slow recruitment. Since Part B1 did not meet its primary objective for confirmed overall response rate, Part B2 (PTEN loss without a PIK3CA mutation) was not initiated, and recruitment was halted for the entire study.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant has a histologically confirmed diagnosis of advanced (loco-regionally recurrent and not amenable to curative therapy), or metastatic (stage IV) TNBC. 2. Participant has adequate tumor tissue to identify the PIK3CA mutation status and the PTEN loss status. 3. Participant has either a measurable disease per RECIST 1.1 criteria or, if no measurable disease is present, at least one predominantly lytic bone lesion or mixed lytic-blastic bone lesion with an identifiable soft tissue component. **Note:** For Part B1, participants must have measurable disease. 4. Participant has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 5. Participant has received no more than one line of therapy for metastatic disease. 6. Participant has adequate bone marrow and organ function.

4.2 Exclusion Criteria 1. Participant has received prior treatment with any PI3K, mTOR or AKT inhibitor. 2. Participant has a known hypersensitivity to alpelisib, nab-paclitaxel or to any of their excipients. 3. Participant has not recovered from all toxicities related to prior anticancer therapies to NCI CTCAE version 4.03 Grade ≤ 1 ; with the exception of alopecia. 4. Participant has central nervous system (CNS) involvement which was not previously treated and/or was newly detected at screening. 5. Participant with an established diagnosis of diabetes mellitus type I or uncontrolled type II based on Fasting Plasma Glucose and HbA1c. 6. Participant has impairment of gastrointestinal (GI) function or GI disease that may significantly alter the absorption of the study drugs (e.g., ulcerative diseases, uncontrolled nausea, vomiting, diarrhea, malabsorption syndrome, or small bowel resection). 7. Participant has a history of acute pancreatitis within 1 year prior to screening or past medical history of chronic pancreatitis. 8. Participant has currently documented pneumonitis/interstitial lung disease. 9. Participant has a history of severe cutaneous reactions, such as Steven-Johnson Syndrome (SJS), erythema multiforme (EM), Toxic Epidermal Necrolysis (TEN) or Drug Reaction with Eosinophilia and Systemic Syndrome (DRESS). 10. Participant with unresolved osteonecrosis of the jaw.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Part A:

Participants randomized 1:1 to receive either alpelisib 300 mg daily orally + nab-paclitaxel 100 mg/m² intravenously (IV) on Days 1, 8, and 15 of each 28-day cycle OR placebo + nab-paclitaxel 100 mg/m² IV on Days 1, 8, and 15 of each 28-day cycle. * **Part B1:** Participants received alpelisib 300 mg daily orally + nab-paclitaxel 100 mg/m² IV on Days 1, 8, and 15 of each 28-day cycle. **Investigational Drugs:** Alpelisib / Placebo (Oral) and paclitaxel / nab-paclitaxel (Intravenous; ATC Code: L01CD01; Trade names: Abraxane, Onxol, Taxol, Paccal). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Supportive care medications (e.g., prophylactic antihistamines to manage rash, medications for glycemic control like metformin, and standard anti-emetics). **Prohibited:** Other concurrent antineoplastic therapies, strong CYP3A4 inhibitors/inducers, and other targeted inhibitors of the PI3K/mTOR/AKT pathway.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor/ctDNA genotyping (PIK3CA/PTEN status), Baseline Imaging
Baseline	Day 0	Randomization, First Dose Administration, Fasting Glucose Assessment
Interim	Every 28-Day Cycle	Intravenous infusions (Days 1, 8, 15), Safety Labs (HbA1c, FPG), Efficacy Assessment via RECIST 1.1, PK sampling
End of Study	Up to ~26 Months	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Progression-Free Survival (PFS) in Study

Part A: Defined as the time from the date of randomization to the date of the first documented progressive disease (PD) or death due to any cause. Assessed via local radiology according to RECIST 1.1 (PD = $\geq 20\%$ increase and absolute increase of ≥ 5 mm in the sum of diameters). 2. **Overall Response Rate (ORR) in Study Part B1:** Based on local radiology assessments in participants with measurable disease at baseline. Defined as the percentage of participants with best overall response of confirmed complete response (CR) or confirmed partial response (PR) according to RECIST 1.1.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) in Study Part A. 2. Overall Response Rate (ORR) with Confirmed Response in Study Part A. 3. Clinical Benefit Rate (CBR) with

Confirmed Response in Study Part A. 4. Time to Response (TTR) and Duration of Response (DOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via ongoing clinical reviews, particularly focusing on known class effects of PI3K inhibitors such as hyperglycemia, severe cutaneous reactions (SJS, TEN, DRESS), diarrhea, and pneumonitis. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours) for events leading to hospitalization, disability, or death. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to review safety and clinical outcome data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy (PFS/ORR); Safety Set for toxicity evaluations. **Handling of Missing Data:** PFS censored at the date of the last adequate tumor assessment if no PFS event was observed prior to the analysis cut-off date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and centralized remote monitoring visits by the sponsor to ensure protocol compliance and patient safety. **Audit Trail:** Validated eCRF completion, central laboratory processing for genomic status, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CBYL719H12301 **Registry Number:** NCT04251533 **Posting ID:** 996660315249619740 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** EPIK-B3 **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Double-blind, Placebo-controlled Study to Assess the Efficacy and Safety of Alpelisib (BYL719) in Combination With Nab-paclitaxel in Patients With Advanced Triple Negative Breast Cancer With Either Phosphoinositide-3-kinase Catalytic Subunit Alpha (PIK3CA) Mutation or Phosphatase and Tensin Homolog Protein (PTEN) Loss Without PIK3CA Mutation **MeSH Code:** D064726 **SNOMED ID:** 709088003

12. Clinical Framework (Oncology Specific)

Primary Condition / Disease Area: Triple Negative Breast Neoplasms (Advanced or Metastatic) **Diagnosis Threshold:** Histologically confirmed TNBC; harboring PIK3CA mutation or PTEN loss **Medical Methodology:** Targeted Kinase Inhibition combined with Taxane Chemotherapy **Optimization Target:** Extension of Progression-Free Survival (PFS) and Objective Tumor Response (ORR)

13. Study Procedures & Methodology

Management Pathway: First- or second-line metastatic systemic therapy **Education Protocol (HCP):** Aggressive monitoring and intervention protocols for alpelisib-induced hyperglycemia and cutaneous reactions. **Education Protocol (Patient):** Counseling on the necessity of fasting glucose checks, strict dietary adherence, and recognizing signs of severe rash/diarrhea. **Quality Audit Mechanism:** Centralized genomic screening to confirm PIK3CA/PTEN biomarker allocation prior to randomization.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity (Follow-up up to ~26 months) **Onsite Visit Cadence:** Weekly infusions (Days 1, 8, 15) for every 28-day cycle **Remote Monitoring Frequency:** Between cycles for safety/toxicity follow-ups **Checklist Review Interval:** Scheduled radiographic tumor assessments (CT/MRI) per RECIST 1.1 at pre-defined intervals

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				